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CHE 515 — Past Questions & Materials
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Course: CHE 515
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 500

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Paper 1 of 2 — CHE 515 CHEMICAL REACTION KINEMATICS ii 2023I2024

Level 500 • First Semester • PastQuestion

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COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

_____ FIRST SEMESTER; 2023/2024 SESSION EXAMINATION —- mipnenanplianiits

- COURSE TITLE: CHEMICAL REACTION KINETICS II Credit units: 3

COURSE CODE: CHE 515 Time Allowed: 3 Hours.

INSTRUCT IONS: Answer any five (5) Questions. All Questions carry equal marks

1a. At 25°C the reaction is strongly exothermic. But this does not interest me because | design to run the reaction at 1025°C. What is the AH, at that temperature and is the reaction still exothermic at that temperature,

The standard heat of gas phase reaction at 25°C is as follows

$A+B \rightarrow 2R$ —Hrassx= -A50,000)

Data Between 25°C and 1025°C the average Cp values for various reaction components are

“Opa =35 J/mol.K Cpp =45J/mol.K —sCpg =70J/mol.K

b. With the aims of a sketched graph, State the effect of temperature on equilibrium conversion as predicted by thermodynamics (pressure fixed). ,

c. Sketch Concentrations of tracer specie > using an impulse inlet.

ae as _ : --

i, PFR(tl=t0+j) wu. CSTR iii. Non ideal reactor

—2a-Determine the equilibrium conversion for-the elementary aqueous-reaction; -between 0°C and 100°C.

A oR AG°29g_ = -14,130 J/mol

AH®°29g = -75,300 J/mol Cen, =Con. *

constant

Present the results in the form of a plot of temperature versus conversion.

b. What restrictions should be placed on the reactor operating isothermally, if we are to obtem a conversion of 75 % or higher.

= ¥ > . > rer

~- a on Pe

3. A sample of the tracer hythane at 320K. was injected as a pulse to a reactor and the effluent concentration was aneasured asa function of time, resulting in the data shown in-tablet below,

The measurements represent the exact concentrations at the times listed and not average values

between the various sampling tests (a) Construct figures showing $C(t)$ and $E(t)$ as function of —
 “_time (b) Determine the fraction of material leaving the reactor that has Spent between 3
 and 6—

minute in the reactor and the fraction of material leaving that has spent between 7.75 ‘and 8.25
 minute in the reactor, And (c) determine the fracture oft material leaving the reactor that has spent
 "3 minute or less in the reactor.

- 4, The catalytic reaction is run at 3.2 atm and 117°C in a plug flow reaction which contain
 0.01kg of catalyst and uses a feed consisting of partially converted product of 20 liter/hr of pure
 unreacted A.

-The results are as follows: | tte —

aC | A Sea aR” |

Cam MoV/Liter Cain Mol/Liter__| 0.100

0.084 000

Find a rate equation to represent i reaction.

5. a. Write short on six (6) general observations made in catalyst activity.

_b, Listed and describe five (5) Spectrum of kinetic Regimes:

c.A——> $4R_-$ -ra= 96Ca mol/kg cat ‘hr

Rage ©F FS, 1mol/iter andé, =3

Consider a catalytic reaction —A—> $4R$, determine the amount S Foatalyst needed in a
 ull

ee bed reactor (assuming eee em OR eR pe ee

——=-movhr-of-pure A-at32-atm-and12C—— - - eS

6a. A pulse of tracer is fed to the reactor and the following exit concentration are reported.

Determine the mean residence time and variance,

Effluent Tracer

the parameter in the formula.

a ee tt et OE TR ae ae eee ~

7, Normal butane C_4H_{10} , is to be isomerized to iso-butane in a plug-flow reactor. Iso-butane is a
 valuable product that is used in the manufacture of gasoline additives. For example, isobutane
 can be further reacts to form iso-octane. The 2004 selling price of n-butane was 72 cents per
 gallon, while the price of isobutane was 89 cents per gallon.

The reaction is to be carry out adiabatically i in the liquid phase under high pressure using —
 essentially trace amounts of a liquid catalyst which gives a specific reaction rate of 31.1h" at
 360K. Calculate the CSTR volumes necessary to process 100,000 gal/day (163 kmol/h) at 70%
 conversion of a mixture 90 mol % n-butane and 10mol % i-pentane, which is considered an inert
 The feed enters at 330K. —--- ania SW Te Irani era Caesar enema me

——— ee

Addit on al information: 3 :

" $A_{K_{irpx}} = -6900 \text{ J/molbutane}$, Activation energy $= 65.7 \text{ TKI/m}^3$

_. $_K = 3.03$ at 60°C — $C_{ao} = -9.3 \text{ Kmol/dm}^3$ Fe

Butane l- Pentane

$C_{pes-B} = 141 \text{ J/mol.K}$ $C_{pi-p} = 161 \text{ J/mol. K}$

$C_{pi4} = 141 \text{ J/mol.K} = 141 \text{ KJ/kmol.K}$

b. In heterogeneous systems, kindly and satisfactorily draw the five ideal flow patterns.

———~¢; Define catalysts and classify catal

ne es re

Paper 2 of 2 — CHE 515 PRESENTATION DOC

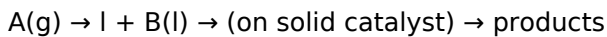
Level 500 • First Semester • Handout

G/L Reactions on Solid Catalysts: Trickle
Beds, Slurry Reactors, and Three-Phase
Fluidized Beds

Multiphase Reactions by Gloria

Multiphase reactions involve reactants in different phases, such as gas, liquid, and solid.

This type of reaction follows this general form:



Here's what happens:

- A (gas) dissolves into the liquid phase.
- B (liquid) reacts with the dissolved A on a solid catalyst.
- The reaction leads to the formation of products.

Types of Reactors Used

This type of reaction can take place in different reactors:

- Packed Bed Contactors: Use large solid catalyst particles.
- Slurry Reactors: Contain very fine suspended solids as catalysts.
- Fluidized Bed Reactors: Can use either large or fine catalyst particles, depending on the flow rates.

Flow Patterns and Their Effect

Flow patterns significantly affect reaction efficiency. The ranking of flow patterns based on driving force effectiveness is:

1. Countercurrent Plug Flow: Most desirable because it provides a higher concentration gradient, improving reaction efficiency.
2. Mixed/Mixed Flow: Least effective because the mixing reduces the concentration gradient, lowering reaction rates.

22.1 THE GENERAL RATE EQUATION by Rukky

To understand the reaction rate, we consider the reaction (1)

The reaction rate equations are:

Where:

$-r_A$ and $-r_B$ are the reaction rates of A and B, respectively, in mol per cubic meter of catalyst per second.

k_A and k_B are the reaction rate constants.

a and b are the concentrations of A and B.

The relationship between rate constants is given by:

$$k_A = k_B / b$$

This equation shows that the reaction rate depends on both reactant concentrations and

the catalyst's effectiveness.

Various ways of running G/L reactions catalyzed by solids by Christian.

This diagram (fig.22.1) shows different ways to mix gas (G) and liquid (L) to carry out chemical reactions using solid materials as catalysts.

There are two main types of setups:

1. Fixed Bed Reactors (A, B, C) – Solid Particles Stay in Place by Magan

Here, the solid catalyst is packed inside a container, and gas and liquid flow through it.

□ (A) Trickle Bed:

- o Liquid slowly drips down through the solid particles.
- o Gas flows through and helps the reaction happen.

□ (B) Bubble Flow in Packed Columns:

- o Liquid fills the spaces between the solid particles.
- o Gas bubbles rise through the liquid and react with it.

□ (C) High-Velocity Flow:

- o Gas is blown in forcefully, pushing the liquid through the solid bed.
- o This speeds up the reaction.

o

2. Reactors with Floating (Suspended) Particles (D, E)

Here, the solid catalyst is not packed in one place. Instead, it is mixed with the liquid to allow better contact with the gas.

□ (D) Agitated Slurry Reactor:

- o Small solid particles are floating in the liquid.
- o Gas bubbles rise through the mixture while everything is stirred.

□ (D) Three-Phase Fluidized Bed:

- o Gas is bubbled through a liquid that has solid particles.
- o The solid moves around freely, allowing better mixing.

□ (E) Slurry Column Reactor:

- o Gas bubbles move up through a liquid with solid particles.
- o This setup makes sure the reaction happens evenly.

How Gas-Liquid Reactions Work on a Solid Catalyst by Donald

This diagram (Fig.22.2) shows how a gas reactant moves through different layers before it reacts on a solid catalyst. The process has multiple barriers (resistances) that slow down the reaction. Understanding these barriers helps us make reactions faster and more efficient.

Gas Moves into the Liquid

- The reactant starts in the gas phase (air or any other gas).

□ It must dissolve into the liquid before reaching the catalyst.

□ First Resistance: The gas has trouble dissolving into the liquid. This is called gas film resistance and is controlled by k_g (gas film coefficient).

□ Example: Imagine trying to mix oil and water—sometimes they don't mix well!

Reactant Travels Through the Liquid

□ Once the reactant dissolves, it moves through the liquid.

□ There is no resistance in the main body of the liquid because the reactant moves freely.

□ But before reaching the catalyst surface, it must pass through a thin layer of liquid around the catalyst.

□ Second Resistance: The reactant moves slowly through this layer, known as liquid film resistance (controlled by k_l).

□ Example: Think of syrup flowing through a straw—it takes effort!

Reactant Reaches the Catalyst Surface

□ Now, the reactant touches the outside of the catalyst.

□ It needs to move inside the tiny pores of the catalyst to react.

□ Third Resistance: The reactant has difficulty entering the pores of the catalyst. This is controlled by k_c (catalyst film coefficient).

□ Example: Imagine water trying to soak into a sponge—it takes time!

Reactant Moves Inside the Catalyst

□ Inside the catalyst, the reactant must travel through tiny tunnels (pores) to reach the active reaction site.

□ Fourth Resistance: If the tunnels are too small, the reactant moves very slowly. This is called diffusion resistance (controlled by D_p , r_p).

□ Example: Think of honey moving through a tiny straw—it's slow!

The Reaction Happens!

□ Once the reactant reaches the center of the catalyst, the chemical reaction takes place.

□ The products then leave the catalyst and go back through the same barriers to exit.

□ Goal: The reaction should be fast and efficient!

Summary Table

Step What Happens? What Slows It Down?

Gas → Liquid Reactant dissolves Gas film resistance (k_g)

Reactant moves in

Liquid → Catalyst Surface Liquid film resistance (k_l)

liquid

Catalyst Surface → Inside Reactant enters pores Catalyst surface resistance ($\frac{1}{k_s}$)

Inside Catalyst → Reaction Diffusion resistance ($\frac{1}{k_d}$, $\frac{1}{k_p}$,

Reactant moves inside

Site $\frac{1}{k_a}$, $\frac{1}{k_d}$)

Graphically we show the resistances as in Fig. 22.2. We can then write the following general rate equations by Daniel:

When two substances react, the speed of the reaction depends on how much of each substance is available and how well they mix.

The reaction rate (how fast the reaction happens) is given by these equations:

For A:

For B:

Now either Eq. 2 or Eq. 3 should give the rate of reaction. Unhappily, even with all the system parameters known (k , a , f , etc.) we still cannot solve these expressions without trial and error because C_A is not known in Eq. 2 and C_B is not known in Eq. 3. However, we usually encounter either of two simplifying extremes, which are very useful.

Extreme Case 1 by Teslim: When B is in Excess ($C_B \gg C_A$.)

Imagine B is a liquid, and there's a lot of it, while A is a gas that dissolves a little bit (slightly soluble) in B. Because there's so much B, its concentration doesn't change much, and we can assume:

$$C_B = C_{B0} \quad \text{everywhere} \quad C_A = C_A \quad \dots \text{same value everywhere}$$

This means that the reaction mainly depends on B, and the equation simplifies to:

Extreme Case 2 by Teslim: When A is in Excess ($C_A \gg C_B$.)

Now, This occurs when B is a liquid and A is a highly soluble gas, often under high pressure. Here, the concentration of A is controlled by how much of it dissolves into B. The reaction mostly depends on A, and another simplified formula is used.

$$C_B = C_{B0}$$

$$C_A = C_A \quad \dots \text{throughout the reactor}$$

□

imagine B is a liquid, and A is a gas that dissolves into B. This time, A is the limiting factor, so its concentration in B controls the reaction. We assume:

How Do We Know Which Case to Use?

Whether these extremes apply.

□ By the unequal signs $\gg$ or $\ll$ we mean two or three times as large:

□ If B is in excess, use Case 1 (Equation 4).

□ If A is in excess, use Case 2 (Equation 5).

22.2 PERFORMANCE EQUATIONS FOR AN EXCESS OF B by Micah, Jonah,

Victory, David, Eghosa. (Micah starts with the intro below and list the various flow patterns)

In reactors like trickle beds, slurry reactors, and fluidized beds, gases and liquids flow through solid catalysts. To analyze how reactions happen, we need to understand the flow of these phases.

Here, we focus on the case where reactant B is in excess. This means:

- The liquid phase (B) is present in large amounts.
- The gas phase (A) is what we need to track.

Mixed Flow G/Any Flow L (Excess of B) by Micah

Gas bubbles move through the solid catalyst bed, as shown in Figure 22.3. This movement creates mixed flow for the gas phase, meaning the gas is evenly distributed throughout the reactor. The gas phase interacts with the catalyst and liquid to drive the reaction.. A material balance on the whole reactor gives:

Overall Reactor Performance:

This equation explains:

- Rate of loss of A $\rightarrow \frac{1}{V} \int_0^V \frac{dA}{dt} dt$ (how much reactant A is used up)
- Rate of loss of B $\rightarrow \frac{1}{V} \int_0^V \frac{dB}{dt} dt$ (how much reactant B is used up, scaled by b)
- Rate of reaction $\rightarrow (-r_A)$ (how fast the reaction happens inside the reactor)

Plug Flow G/Any Flow L (Excess of B) by Jonah

Imagine bubbles of gas rising through a liquid in a reactor as shown in Fig. 22.4. The gas contains A, which reacts with B in the liquid.

- B is in excess, meaning there is so much B that its concentration doesn't change much.
- The amount of A in the gas changes as it reacts and moves upward.
- The gas enters with a certain concentration of A (A_0) and leaves with a lower concentration (A).
- Since B is in excess, we assume r_A is constant everywhere.

Overall Reactor Performance:

Mixed Flow G/Batch L (excess of B) by Victory:

We have gas (G) bubbling through a batch of liquid (B). The gas reacts with a specific component dissolved in the liquid. Since liquid B is in excess, its concentration does not change significantly, while the gas is gradually consumed.

Imagine a tank filled with liquid B where gas bubbles are constantly rising. As the gas dissolves in the liquid, it undergoes a chemical reaction. Because there is a large amount of liquid, the overall composition of B remains almost constant, while the gas concentration decreases over time as seen in Fig. 22.5.

Here the material balance at any time t becomes

Then from II and III or I and II solve for t

Plug Flow G/Batch L (excess of B) by David:

This section explains how gas moves through a batch of liquid in a plug flow manner, with an excess of reactant B present.

In this system:

- Gas flows continuously through a stationary liquid.
- The gas composition changes gradually over time as it moves.
- The liquid phase is in a batch mode, meaning it does not move.

A plug flow means that the gas moves without mixing in the flow direction, meaning each gas element experiences the same conditions as it travels.

A material balance equation accounts for the amount of reactant A that reacts:

$$F_{A0} - F_A = (-r_A) V \quad (13)$$

Finding the Exit Conversion of A; Integrating the above equation gives:

This tells us how much reactor volume is required to achieve a certain conversion of reactant A.

Considering the Effect of Excess B: Since B is in excess, it does not limit the reaction, but it affects how A reacts.

We introduce another equation considering the change in concentration of B over time:

To find how long the process takes, determining the process time, we integrate:

Where;

- t = Total processing time (s)
- V = Volume of liquid in the batch (L)
- ν_B = Stoichiometric coefficient of B in the reaction
- F_{A0} = Initial molar flow rate of reactant A (mol/s)
- C_B = Concentration of reactant B in the liquid (mol/L)
- C_{B0} = Initial concentration of B (mol/L)
- C_B = Final concentration of B after reaction (mol/L)
- X_A = Exit conversion of A (fraction reacted)

Special Case of Pure Gaseous A (excess of B) by Eghosa:

This section deals with a common scenario in chemical reactions, particularly hydrogenation reactions, where pure gaseous A is involved, and liquid B is in excess.

Why Does This Matter?

- In many cases, A is a gas (e.g., hydrogen in hydrogenation reactions), and B is a liquid.
- The gas (A) is often recycled, meaning its pressure P and concentration C_A stay nearly constant.
- This simplification allows us to use easier mathematical equations for both batch and flow reactor systems.

The figure (Fig. 22.9) shows a setup where pure gaseous A can be recompressed and recycled after reaction.

So far, we've assumed first-order reactions:

□ First order with respect to A.

□ First order with respect to B.

However, what if the reaction does not follow first-order kinetics for both reactants? We need a more general rate equation.

Where the reaction rate is given as:

where:

□ □□ = rate constant

□ □□ = concentration of A

□ □□ = concentration of B

□ □, □ = reaction orders with respect to A and B

This equation is more complex than the simple first-order assumption.

To make calculations easier, we approximate the rate equation using a first-order form:

Why This Approximation?

□ Instead of dealing with complicated rate orders n and m, we approximate it using a first-order reaction in □□.

□ This is done by treating (□□□ □□□) □□□−1 as a constant at the locations where the reaction occurs.

To accurately describe the overall reaction rate, we must consider mass transfer limitations. The complete rate expression now becomes:

22.3 PERFORMANCE EQUATIONS FOR AN EXCESS OF A by Pius

When there is an excess of reactant A, its concentration remains nearly constant, meaning we only need to focus on the behavior of reactant B in the liquid phase. The equations below describe how different reactor types perform under these conditions.

Plug Flow L/Any Flow G (Tower and Packed Bed Operations)

For plug flow systems, including towers and packed bed reactors, the performance equation is obtained by integrating the material balance:

The conversion □□ can be related to the concentration of B, where □□ is the concentration of B at any point, and □□0 is the initial concentration of B. This equation is crucial for designing plug flow reactors based on conversion and reaction rate.

Mixed Flow L/Any Flow G (Tank Operations of All Types)

For mixed flow or continuous stirred-tank reactors (CSTR), the performance equation simplifies to:

This equation helps determine the required reactor volume based on the reaction rate and conversion level in a CSTR.

Batch L/Any Flow G

For batch reactors, since reactant A is in excess, its concentration is assumed constant, leading to the following rate equation:

Rearranging and integrating over time:

V is the volume of liquid in the batch reactor

t is the reaction time

This equation helps calculate the time required for the batch reaction to reach a certain concentration.

22.4 WHICH KIND OF CONTACTOR TO USE by Sarah and Magan (Sarah starts first)

The selection of a good contactor depends on

- where the controlling resistance lies in the rate expression,
- the advantages of one contacting pattern over another, and
- the difference in auxiliary equipment needed.

The overall economics which accounts for these three factors will determine which set up and reactor type is best. Let us briefly look at these factors in turn.

The Rate: We should favor the contactor which favors the weakest step in the rate. For example,

- if the main resistance lies in the G/L film, use a contactor with large interfacial surface area.
- if the resistance lies at the L/S boundary, use a large exterior surface of solid, thus large ϕ or small particles.
- if the resistance to pore diffusion intrudes, use tiny particles.

From prediction we can find the weakest step by inserting all transfer coefficients ($k_{G/L}$, $k_{L/S}$, . . .) and system parameters (ϕ , ρ , . . .) into the rate equation and then see which resistance term dominates. Unfortunately, the values for these quantities usually are not well known.

Magan continues: From experiment, we can change one or another factor in the rate expression; for example,

- solid loading (this changes ϕ alone, hence, only changes the last resistance term in the rate expression)
- size of catalyst particle (affects both ϕ and ρ)
- intensity of agitation of the liquid (affects the mass transfer terms in the rate)
- $k_{G/L}$, $k_{L/S}$, k_p .

This should tell which factor strongly affects the rate and which does not.

Boosting the weakest step in the rate by a proper choice of particle size, solid loading and reactor type can strongly affect the overall economics of the process.

Contacting: Plug flow of the limiting component, the one which is not in excess, is certainly better than mixed flow. However, except for very high conversions this factor is of minor importance.

Supporting equipment: Slurry reactors can use very fine catalyst particles, and this can lead to problems of separating catalyst from liquid. Trickle beds don't

have this problem, and this is the big advantage of trickle beds. Unfortunately, these large particles in trickle beds mean much lower reaction rates. With regard to rate, the trickle bed can only hold its own

- for very slow reactions on porous solids where pore diffusion limitations do not appear, even for large particles and

- for very fast reactions on nonporous catalyst-coated particles.

Overall, the trickle bed is simpler, the slurry reactor usually has a higher rate, and the fluidized bed is somewhere in between.

22.5 APPLICATIONS by Anita

Here is a short list of applications of these reactors.

- The catalytic hydrogenation of petroleum fractions to remove sulfur impurities. Hydrogen is very soluble in the liquid; high pressure is used while the impurity is present in the liquid in low concentration. All these factors tend to lead to extreme 2 (excess of A).

- The catalytic oxidation of liquid hydrocarbons with air or oxygen. Since oxygen is not very soluble in the liquid, while the hydrocarbon could well be present in high concentration we could end up in extreme 1 (excess of B).

- The removal of dissolved organics from industrial waste water by catalytic oxidation as an alternative to bio-oxidation. Here oxygen is not very soluble in water, but the organic waste is also present in low concentration. It is therefore not clear in what regime the kinetics lie. The catalytic oxidation of phenol is an example of such an operation.

- The removal of airborne pollutants by adsorption and/or reaction. These operations usually lead to extreme 1 (excess of B).

- The illustrative examples and many of the problems which follow are adapted from or are extensions of problems prepared by Ramachandran and Choudhary (1980).

Calculations by Maroke